
Instructions. (15 points) Sections 15.7, 15.8, 16.1, 16.2. Show relevant work for full credit.

- (4^{pts}) **1.** Evaluate the integral $\iiint_E z\sqrt{x^2 + y^2} dV$, where E is the solid inside the cylinder $x^2 + y^2 = 1$, between $z = 0$ and the cone $z = \sqrt{x^2 + y^2}$.

- (4^{pts}) **2.** Evaluate the integral $\iiint_E z dV$, where E is the region inside the sphere $x^2 + y^2 + z^2 = 4$ and above the cone $\phi = \pi/3$.

(3^{pts}) **3.** (a) (2 pts) Compute ∇f for $f(x, y) = xe^y + y^2$.

(b) (1 pts) Using part (a), explain why the vector field $\mathbf{F}(x, y) = \langle e^y, xe^y + 2y \rangle$ is conservative.

(4^{pts}) **4.** Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = \langle x + y, y \rangle$ and C is the line segment from $(1, 0)$ to $(3, 2)$.